5 Fig. 5.1 shows a circuit containing a capacitor of capacitance C and a resistor of resistance R.

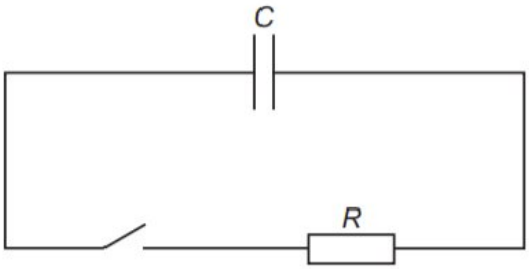


Fig. 5.1

Initially, the switch is open and the potential difference (p.d.) across the capacitor is 12 V.

The switch is closed at time $t = 0$ and the capacitor discharges through the resistor.

Fig. 5.2 shows the variation of the charge Q on the capacitor with the p.d. V_C across the capacitor as the capacitor discharges. Fig. 5.3 shows the variation of the current I in the resistor with the p.d. V_R across the resistor as the capacitor discharges.

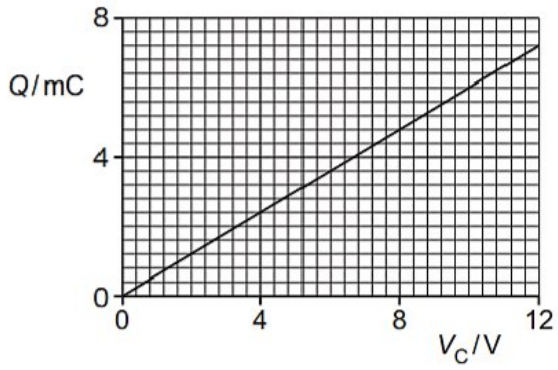


Fig. 5.2

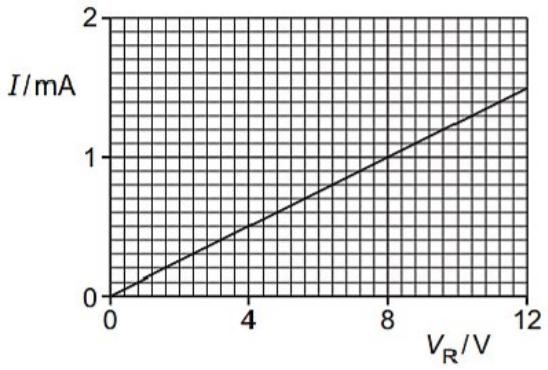


Fig. 5.3

(a) Determine

(i) the capacitance C,

C = ... F [1]

(ii) the resistance R,

R = ... Ω [1]

(iii) the time constant τ.

τ = ... s [1]

(b) A quantity is said to vary exponentially when its rate of change is proportional to the quantity itself.

Using Fig. 5.2, Fig. 5.3 and the relationship between V_C and V_R, explain why the variation of Q with t is exponential in nature.

.....

.....

.....

..... [2]

(c) **(i)** The potential difference before the capacitor starts discharging is V_O. Determine, in terms of V_O, the potential difference across the capacitor when the energy stored in the capacitor is half its initial value.

potential difference = ... [2]

(ii) The time taken for the potential difference of the capacitor to be reduced to half its initial value is T. Determine, in terms of T, the time taken for the energy stored in the capacitor to decrease to half its initial value.

time taken = ... [2]

[Total: 9]